

**Analysis on Vacuum O-Ring Seals and its Material Selection via
Granta Software**

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1. Introduction

High-vacuum systems constitute an essential cornerstone for modern technological advancements in industries such as aerospace engineering, semiconductor manufacturing, and electron microscopy. In these fields, vacuum seals are conventionally categorized based on their target operational pressure ranges—namely rough vacuum (10^{-3} Torr), High Vacuum (HV) (10^{-5} to 10^{-8} Torr), and Ultra-High Vacuum (UHV) (10^{-10} to 10^{-11} Torr)—as well as their thermal resilience during high-temperature bake-out procedures [1]. However, achieving and sustaining this level of isolation becomes significantly more complex in instruments that require mechanical motion, where dynamic vacuum seals must withstand both demanding mechanical wear and chemical exposure.

To overcome these stringent demands, high-performance vacuum O-rings are heavily utilized due to their exceptional elastic recovery. These toroidal devices are conventionally seated within a machined metal groove or gland, where they deform elastically upon installation [2]. This deformation generates the compressive stress necessary to achieve a reliable seal; the material microscopically conforms to and fills any surface imperfections along the mating interfaces. Consequently, this continuous contact blocks the ingress of atmospheric gases into the vacuum chamber, thereby guaranteeing the long-term stability of the controlled environment.

Driven by these critical operational and structural requirements, the present document focuses on conducting an appropriate material selection for a high-performance Vacuum O-ring specifically deployed within a Scanning Electron Microscope (SEM) environment. By evaluating and optimizing key polymer properties against the strict demands of high-vacuum sealing, this analysis aims to identify the most reliable solution to ensure system integrity and prevent vacuum degradation.

2. Application for a Scanning Electron Microscopy (SEM) Device

As previously noted, the field of electron microscopy represents one of the primary applications for high-vacuum O-rings, with the Scanning Electron Microscope (SEM) serving as a prime example. This high-resolution instrument focuses an electron beam onto a sample to generate detailed topographic images [3]. To prevent electron beam scattering and specimen contamination, the SEM chamber must maintain a High Vacuum (HV) baseline pressure between **10^{-5} and 10^{-6} Torr**.

Our analysis models the architecture of standard commercial SEMs from leading manufacturers such as Thermo Fisher Scientific, JEOL, and Zeiss. Specifically, our target interface is the large-perimeter access door of the specimen chamber when operating between the range of room temperature and **$\sim 150^{\circ}\text{C}$** (No cryogenic scenario contemplated). This heavy drawer is frequently cycled (opened to the laboratory atmosphere for sample loading and then sealed for rapid pump-down). So, a **dynamic operation** is the one contemplated for the analysis.

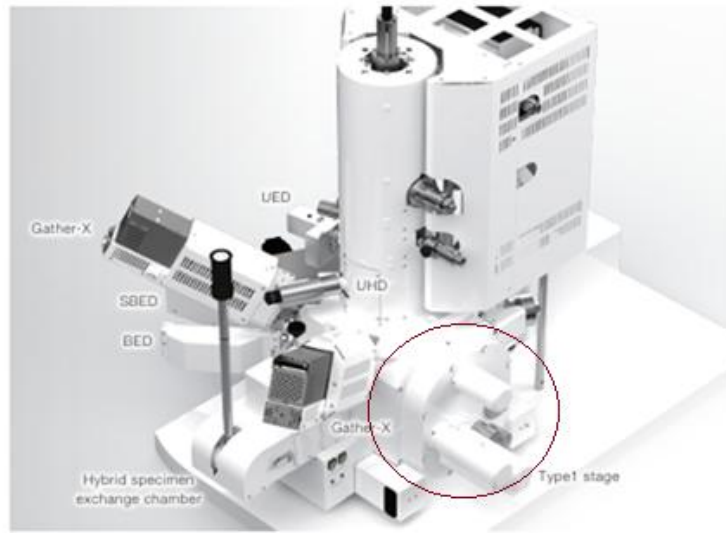


Figure 1. Scanning Electron Microscope, O-ring placement highlighted in red [4].

3. Project Methodology

The primary objective of this case study is to categorically define an optimal material for the dynamic seal of the SEM specimen exchange door, modelled upon the data provided by one of the leading OEMs of the world. The specific interface would have to face a multidimensional stress profile that the standard commercial rubbers cannot withstand. The selected material should also be able to prevent severe contamination by trying to exhibit near-zero outgassing of the Volatile Condensable materials (VCMs) under an electron beam, while also trying to maintain structural stability across bake-out cycles, vacuum loading/unloading during specimen exchange and constant environmental exposures (*ASTM E1559*). Also, the material must possess a critically low compression set to guarantee instantaneous elastic recovery, ensuring rapid chamber pump-down times without the risk of virtual leaks (*ASTM D395 and ISO 815 testing methodology*).

To resolve these rigorous mechanical, chemical and environmental constraints, the project employs the Ashbey material selection methodology. By translating the dynamic operational conditions of the SEM chamber into quantifiable indices, we will utilize Granta EduPack software to systematically filter out and in the end, isolate the specialized materials which will then be analyzed for the ecological and sustainability constraints.

4. Design Considerations

For our selection process we must set the main indexes, limitations and parameters that are essential in order to pick the right material for the application. For this, a literature review of the actual international standards, documents for high performative Vacuum O-Rings design and papers on the topic will be conducted to evaluate different approaches on this matter.

As an initial boundary condition for our research, the scope of this material selection has been restricted to **polymers**. Evaluation of the state of art and technical literature confirms that these materials provide the best compliance, gas permeability resistance, and sealing efficiency demanded by vacuum applications. Metal lacks the elasticity to recover from continuous cycling, leading to permanent deformation, fatigue cracking, and high-friction wear, whereas polymers flexibly adapt without breaking. Therefore, to optimize the evaluation process, non-polymeric materials have been screened out, and further analysis will strictly focus on optimizing polymer selection.

4.1 O-Ring Architecture

To satisfy the rigorous constraints of the SEM chamber, the dynamic seal should be structurally defined prior to material screening stage in Granta EduPack afterwards. In high performance vacuum sealing applications, there are usually two primary architectural configurations: the monolithic (uni-material) approach and the composite (encapsulated or jacketed O-ring) approach.

- **The Monolithic (Homogenous) Seal:** This configuration opts for an "unibody" design using a single advanced material to ensure structural predictability, allowing its performance to be directly validated against established NASA, ASTM, and ISO vacuum standards. While this uniform polymer matrix simplifies the design process by leveraging reliable literature constraints, its primary drawback is the high financial cost: finding a single material whose intrinsic chemistry can simultaneously satisfy extreme mechanical elasticity (low compression set), resist chemical degradation, and minimize outgassing rates typically demands expensively engineered and specialized materials.



Figure 2. Monolithic O-ring (Cross-sectional view)

- **The Composite (Encapsulated) Seal:** Alternatively, this multibody architecture uses an encapsulated seal (an elastic core inside a chemically inert PTFE/FEP jacket) to decouple engineering requirements, using the core for elasticity and the outer layer for low outgassing and chemical resistance [7].



Figure 3. Encapsulated O-ring (Cross-sectional view)

4.2 Geometry Characteristics & Sizing

The dimensions of the O-ring are essential to define the compressive setting, area of overage, permeation surface area and the probable thermal expansion in our settings.

According to the OEM configuration data [4], the Type 2 and 3 eucentric goniometer stages must accommodate an allowable specimen diameter of 200 mm. Thereby, to provide an adequate physical clearance for the operator during the specimen exchange, the exchange chamber's access area port should be larger than the allowable specimen size and thereby, a nominal port flange diameter of 250mm is idealized.

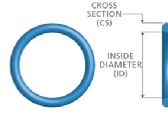
In critical operations like vacuum engineering, custom O-ring sizes are usually abstained from, to reduce the risk of any catastrophe or failure. So, internationally standardized dimensions are utilized to ensure reliable groove seat machining and repeatable compression. Pertaining to our selection, *SAE AS568 (Aerospace Size standards for O-rings)* is consulted that operates in conjunction with *ISO-3601-1* metric standard (*O-rings for Fluid Power Systems*).

Looking at the specification chart for SEM, the O-ring AS568-377 is selected for our application as in accordance with the industrial standards, it will provide the required circumferential profile to accommodate the connection flange surface imperfections and being thin enough to be compressed by the installed latching mechanisms.

The standardized dimensions of the AS568-377 O-ring are.

- Inner Diameter (ϕ_{IN}): 253.57mm
- Cross-sectional Diameter (ϕ_{CS}): 5.33mm
- Mean Diameter (ϕ_M): 258.9mm

Table 1. Standard Dimension Table for AS568-377 O-ring



Dash #	CS nominale	ID nominale	OD nominale	CS Inch	ID pollici	Tol Inch	CS mm	ID mm	Tol mm
377	3/16	10	10-3/8	0.21	9.975	± .055	5.33	253.37	± 1.40

From this dimensional data, the variables required for mechanical, chemical and permeation calculations are derived.

The Perimeter (**P**) of the O-ring can be provided as:

$$P = \pi(\phi_M) = \pi(258.99 \times 10^{-3}) \approx 0.819\text{mm} \quad (1)$$

Assuming that upon installation, half of the surface area of the O-ring would be exposed to the internal vacuum chamber conditions, the permeation and outgassing surface area can be calculated as:

$$A_{s(per)} = \frac{\pi \cdot \phi_{CS} \cdot P}{2} = \frac{\pi(5.33 \cdot 10^{-3})(0.819 \cdot 10^{-3})}{2} = 0.0068 \text{ m}^2 \quad (2)$$

4.3 Thermal Boundary Conditions

Before the calculation of thermal expansion and compression set degradation, the absolute thermal limits of the system must be specified. The O-ring acts as a thermal barrier between the outside laboratory environment and the internal vacuum chamber, thereby resulting in exposure to dual thermal profiles: steady-state operations and extreme decontamination cycles.

4.3.1 Steady-State Operation

According to the provided OEM data (*Table: Installation requirements*) [4], the SEM requires a controlled external laboratory environment.

- **Installation Room Temperature:** $20 \pm 5^\circ\text{C}$
- **Cooling Water Temperature:** $20 \pm 5^\circ\text{C}$

This provides the baseline requirement. So, under standard imaging procedures, the active water-circulation system maintains the chamber walls at ambient temperature and thus the steady-state operational temperature of the O-ring can be considered as 20°C .

4.3.2 Internal Chamber Conditions (Vacuum-Bakeout)

For our application, the primary thermal constraint doesn't arise from the standard operation but from the maintenance of vacuum within the chamber. So, during the unloading/loading event, the atmospheric moisture gets adsorbed on the internal chamber walls. To reach the operational baseline pressure of 10^{-5} Pa, these vapors are required to be desorbed. This is achieved through the bake-out procedure, when the internal heaters raise the chamber temperature and the Turbomolecular pumps evacuate the outgassed moisture.

From the OEM datasheet, the HV chamber bake-out temperatures lie within the range of 100-150 °C. Furthermore, from the aerospace outgassing standards specified in *ASTM E595 [5]*, the specimens were subjected to vacuum at the temperature of **125 °C** for 24 hours. So, to define a safe margin keeping in view both the bake-out protocols and the available standard procedures, the upper thermal boundary limit is defined at **150°C**.

So, the maximum thermal gradient (ΔT) = **150 – 20 = 130°C** defines two absolute constraints for the selection phase.

The selected material must possess a continuous service temperature ceiling greater than **150 °C** to avoid melting, matrix degradation and adhesion to metallic flange. At the same time, the glass temperature (T_g) of the polymeric material (usually employed in industrial applications for O-rings) should be higher than **150°C** to ensure elasticity and proper specimen encompassing without brittle failure specifically during cold starts.

4.4 Thermal Expansion & Strain Compatibility

As already specified, during the bake-out procedure, the dynamic seals transition from the ambient 20°C to a maximum thermal gradient of 130°C. As learnt from the theoretical lectures during our course, the elastomers usually possess a higher thermal coefficient of expansion (α) in comparison to the metals (that is utilized in the flange of metallic chamber). This thermal cycle is going to introduce severe dimensional changes which the target material should be constrained for to avoid extrusion or internal rupture.

4.4.1 Monolithic Approach

In the unibody seal design, the primary risk related to thermal expansion is the *gland overfill*. According to the aerospace grade sealing requirements (highlighted in *Parker O-ring Handbook ORD 5700 [6]*), the groove machined into the SEM flange is designed with the specific void volume to accommodate for seal deformation. If the volumetric expansion of the material exceeds beyond the available void space, the material would behave as an incompressible fluid and will exert the hydrostatic pressures to such extent that fractures the metal groove or extrudes the sealing into the vacuum chamber.

To ensure that the O-ring does not exceed the maximum gland fill limit of 90%, the volumetric expansion ($\frac{\Delta V}{V}$) should be bound tightly. As for the usual industrial standards, the polymers are utilized

for these operations, and their volumetric expansion is approximately three times the linear thermal expansion. So, the linear CTE of the target material can be defined by:

$$\left(\frac{\Delta V}{V}\right) = 3(\alpha)(\Delta T) \quad (3)$$

By setting a safe upper limit on the volumetric expansion (typically <10% for high-temperature static hardware), it becomes a critical filtering parameter in Granta EduPack.

4.4.2 Composite Approach

For the multi-material architecture, the outer jacket should be composed of a non-porous fluoropolymer (such as PTFE or FEP) to provide the necessary chemical inertness and vacuum barrier. The standard α_{PTFE} is known and is around $120 \times 10^{-6} K^{-1}$.

So, the unknown variable is the internal elastic core which depends on the type of composite O-ring selected. As the core and jacket are physically bonded, the difference between their CTEs would induce internal shear strains ($\Delta\epsilon$). This mismatch can be calculated by.

$$\Delta\epsilon = (\alpha_{\text{core}} - \alpha_{\text{jacket}})\Delta T \quad (4)$$

At elevated temperatures, the fluoropolymer jackets are susceptible to micro-fracturing if the applied tensile strains exceed beyond 2% i.e., $\epsilon_{\text{max}} = 0.02$ (it is because PTFE and FEP do not possess elastic memory beyond 2-3% elongation and then undergo permanent deformation which would be dangerous for our dynamic operation [7]). So, to prevent the core from expanding and rupturing the jacket during the exposure to thermal gradient, the CTE of the core should be mathematically constrained.

$$0.02 \geq (\alpha_{\text{core}} - 120 \times 10^{-5})(130)$$

$$\alpha_{\text{core}} \leq \frac{0.02}{130} + 120 \times 10^{-5}$$

$$\alpha_{\text{core}} \leq 273 \times 10^{-6} K^{-1}$$

So, if the composite architecture is selected. The identified core material should possess a linear CTE less than $273 \times 10^{-6} K^{-1}$ to prevent the delamination of vacuum barrier during bake-outs.

4.5 Pressure Gradient Limits

As already mentioned, the dynamic seal of the specimen exchange chamber in the SEM will not operate under a static load but every time during specimen exchange or observation, loading/unloading process exposes it to cyclical pressure gradients and rapid pump-downs. So, the target material should be mechanically able to withstand these transitions without structural yielding, extrusion or fatigue.

According to the reference OEM specifications [4], the chamber is cyclically vented to the standard atmospheric pressure (101.325 kPa) during the specimen exchange procedure, usually along with the introduction of dry Nitrogen gas. After the exchange is completed, the roughing pumps and Turbomolecular Pumps (TMP) rapidly evacuate the chamber to High-Vacuum (HV) baseline of 10^{-5} Torr ($\approx 1.33 \times 10^{-3}$ Pa).

For the sake of simplicity, to establish the gradient while keeping the calculations simple, the internal vacuum pressure is absolute zero. Thereby, during normal steady state operation, the pressure differential (ΔP) of 101.325 kPa is established.

This pressure differential creates a massive hydrostatic force to crush the door into the vacuum chamber. As, from section 4.2, the maximum idealized door diameter was specified to be 250mm. So, keeping it as the baseline, the clamping force (F_{atm}) acting on the door assembly can be calculated as:

$$F_{atm} = (\Delta P) \cdot \pi \cdot \left(\frac{\phi_{IN}}{2}\right)^2 = (101.325 \times 10^{-3})(\pi) \left(\frac{250 \times 10^{-3}}{2}\right)^2 \approx 4,974 \text{ N}$$

Upon installation, in accordance with the standard vacuum sealing guidelines for hydraulic and pneumatic applications (*ISO 3601-2 [8]*), the standard is specifically characterized for hydraulic and fluid power applications. Sealing against highly penetrative atmospheric gases in a high vacuum (HV) at 10^{-5} Torr requires a significantly higher localized contact stress to seal the metal asperities of the flange. So, also referring to the *Parker ORD 5700 O-ring guide*, the specified guidelines for aerospace applications dictate a 25% compression of the cross-section for the initial seal.

In our case, the 5.33mm AS568-377 O-ring has a flattened contact width (w) of approximately 1.33 mm. The resultant contact area (A_c) along the perimeter (P) is:

$$A_c = P \cdot w = (0.813)(1.33) = 0.00108 m^2$$

From this, we can determine the compressive stress (σ_c) exerted on the material as:

$$\sigma_c = \frac{F_{atm}}{A_c} = \frac{4,974}{0.00108} = 4.60 \text{ MPa}$$

So, we can conclude that the target material should possess the target compressive yield strength of more than **4.60 MPa** to avoid structural yielding.

4.5.1 FEA Validation

4.5.1.1 Analytical Assumptions & Methodology

The validation of these results was carried out to validate the empirical analytical calculations against a non-linear hyperelastic FEA. The calculations estimated a required compression force of 4974 N to achieve a 1.066 mm displacement (compression), assuming a continuous 3D toroidal geometry that involves hoop stress. To optimize the computational resources, the FEA study utilized a 2D Plane Strain assumption with a Mooney-Rivlin material definition (as its required for SolidWorks analysis, so a custom elastomer was selected for evaluation), thereby modelling the seal as an extruded linear cross-section. A symmetry boundary condition was applied to the vertical centerline. This stabilized the frictionless contact simulation against the rigid body motion (zero pivot errors) while permitting natural, unconstrained lateral bulging of the elastomer within the flange.

4.5.1.2 Results & Discrepancy Analysis

The simulation resolved to 100% displacement, yielding a peak Y-reaction force of 2771 N and a symmetric Hertzian peak contact pressure of 1.6 MPa. These results are very different from our analytical results (4974N and 4.6 MPa). It can be explained from the 2D plane strain assumption which isolates the cross-sectional deformation but ignores the 3D hoop stiffening. Furthermore, empirical equations typically compute the generalized internal stress, whereas the non-linear FEA model calculated the true localized surface interactions. So, to guarantee a robust hermetic seal, the higher analytical values will be adopted as the conservative design baseline as it provides a natural factor of safety to overcome the unmodeled real-world variables such as 3D hoop stiffening, surface friction and batch variations.

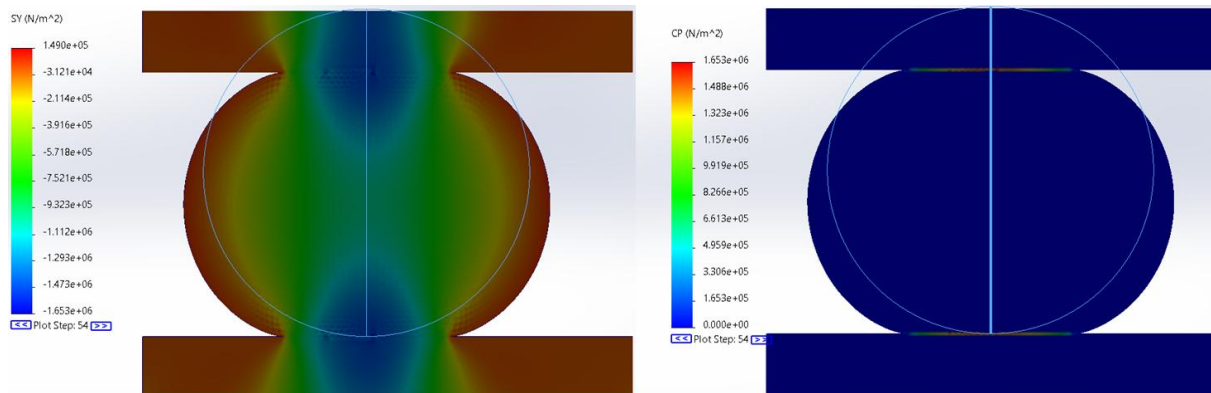


Figure 4. FEA results (2D cross-section in between two plates (depicting a flange)). Normal Stress (y-axis) (Left) and Contact pressure (CP) (Right)

4.5.2 Monolithic Approach

Monolithic elastomers typically exhibit a hyper-elastic behavior i.e.; they can undergo 25% cross-sectional deformation while being within the elastic limits. So, ideally during the loading/unloading of the specimen within SEM, the monolithic design fully recovers its toroidal geometry, thereby countering cycle fatigue for repeated specimen exchanges over multiple cycles during the operational life of SEM.

4.5.3 Composite Approach

While the inner core can mimic the properties of monolithic design, the outer thermoplastic jacket (PTFE or FEP) dictates the physical surface contact. Thermoplastics possess significantly lower elastic strain limits as compared to elastomers and thereby undergoing compressive squeeze to achieve the target 4.60 MPa sealing stress, the jacket will be forced beyond its yield point. This would cause permanent deformation (creep or cold-flow) every cycle which will accumulate over continuous operation, ultimately resulting in failure of the sealing interface.

4.6 Surface Hardness & Indentation Resistance

The gas permeation across the metal-polymer interface is governed by the surface hardness and plays a crucial role in setting up constraints to avoid too much outgassing during the specimen exchange. The target material is required to be soft enough to elastically deform and cover up the microscopic machining grooves while (asperities) of the SEM flange, while being hard enough to resist physical extrusion into the diametral clearance gap under the steady state pressure differential.

From ASTM D2240, the hardness for thermoplastics and elastomers is quantified using the Shore A durometer scale. Again, with reference from Parker O-ring guide, the optimal hardness for a dynamic seal is strictly bound between 70 and 80 Shore A.

4.6.1 Monolithic Approach

Materials with the shore hardness lower than 70 are highly susceptible to vacuum extrusion and abrasive tearing during the dynamic operation cycle. However, elastomers can also be molded to hit the exact hardness values of 70-80 Shore A, allowing them to cover the microscopic asperities while keeping their mechanical elasticity.

4.6.2 Composite Approach

On the other hand, the materials with shore hardness A higher than 85, or the composite seals that employ rigid thermoplastic jackets are measured on the much harder shore D scale pose a significant operational risk. Seals with high hardness require a massive clamping force to conform the deal to the flange, which can pose a risk of warping of SEM's chamber opening and door hinges overtime. Also, the rigid outer coating lacks the required elastic memory and thereby if a microscopic dust particle or contaminant gets entrapped against the flange, the hard surface might face a permanent plastic indentation, resultantly generating an irreversible leak path across the seal face.

Thereby, to avoid vacuum extrusion, micro-scale conformability and indentation damage during operation, the material selection profile in Granta EduPack would be bonded to elastomers with the hardness ranging from 70 to 85 Shore A in accordance with the literature review explained.

4.7 Dynamic Elastic Recovery (Compression Set)

During the sample exchange, when the clamping force is removed, the O-ring must undergo instantaneous elastic recovery to bridge back the flange gap. If the material gets permanently deformed (under constant clamping load, adjusts and deforms according to the clamping force and undergoes “relaxation” losing its ability from elastic deformation to permanent strain), the seal will fail to rebound and create leak points or microscopic gaps that would propagate and grow during each pump-down cycle.

This inability to elastically recover is termed as Compression set. In industrial settings, the *ASTM D395 (Method B – Compression set under Constant Deflection)* [9] is referred to for aerospace and vacuum applications. This involves compressing the specimen to specific fraction of its original thickness and evaluating the permanent deformation after exposure to elevated temperatures.

The Compression set (C_B) is given by the formula...

$$C_B = \frac{t_o - t_i}{t_o - t_n} \times 100\% \quad (5)$$

Applying for our reference O-ring size AS568-377 Seal, the geometrical parameters can be defined as:

t_0 (Original Thickness): 5.33 mm

t_n (Compressed Thickness): Based on the 25% static vacuum shrinkage, the groove spacer thickness is $5.33 \times (1 - 0.25) \approx 4$ mm.

t_i (Recovered Thickness): The resultant thickness of the specimen after the loading is removed.

In standard hydraulic applications, the compression set of up to 40% is acceptable. However, in our case of SEM based vacuum application, the environment plays a critical role i.e., 150°C thermal bake out. These temperatures accelerate the visco-elastic relaxation of polymer chains (mostly employed industrially for these applications-taken as reference).

Thereby, we have adopted the parameters defined by extreme environmental conditions i.e., Aerospace industries where they utilize *Aerospace Material Specifications (such as SAE AMS 3216)* and *Parker O-ring Handbook 5700* for relevant data, both of which confer to high-temperature, high vacuum elastomers under specific sections.

According to these standards, to prevent the loss of dynamic sealing force and avoid virtual leaks, the maximum permanent deformation must be strictly bounded. They refer to *ASTM D395* testing at 200°C for 70 hours and propose a compression set of **20%**. Defining this as our baseline criterion, we can formulate equation (5) as:

$$20\% = \frac{5.33 - t_i}{5.33 - 4.00} \times 100\%$$

$$0.20 = \frac{5.33 - t_i}{1.33}$$

$$t_i = 5.33 - 0.266 = 5.064 \text{ mm}$$

So, the O-ring should recover to 5.064mm thickness after every cycle (even after prolonged 150°C bake-outs) to maintain a functional dynamic seal.

4.7.1 Architecture Considerations

As dictated from the material property charts, the monolithic elastomers reliably recover beyond the stated threshold, thereby allowing the sealing reset for the next cycle.

However, encapsulated seals are incompatible with this constraint due to the phenomenon of thermoplastic creep (cold flow). Even though the inner elastomeric core retains its elastic memory, the outer PTFE/FEP jacket undergoes severe plastic flow due to the thermal bake-out and clamping force, causing an inability to recover within the allowable limits. This inability translates further each cycle causing the seal to fail under dynamic operational requirements.

4.8 Steady State Gas Permeation

Most of the materials, including the polymers employed in the vacuum applications, are microscopically porous to atmospheric gases (N₂, O₂ and H₂O-water vapors). The continuous exchange of these elements from the laboratory environment to the high-vacuum chamber presents a “leak” in the system which dictates the ultimate pressure limit for the SEM.

According to **Henry’s Law**, the concentration of dissolved gas (c) is proportional to the external pressure (P) and the specific solubility of the material (s). This can be defined as:

$$c = P \cdot s$$

Subsequently, the dissolved gas migrates through the polymer matrix. For the cylindrical geometry of an O-ring, the mass flow (j) is analytically provided by **Fick's First law of diffusion**...

$$j = D \frac{\partial c}{\partial r}$$

Where D is the diffusion coefficient and r is the radial distance.

In standard vacuum engineering practices, these properties are combined into the Permeation Constant (K) which can be defined as.

$$K = D \cdot s$$

Applying this to the calculated geometry of our O-ring, the total steady-state volumetric flowrate (Q) entering the SEM is.

$$Q = \frac{K \cdot A_s \cdot \Delta P}{w}$$

(Where $A_s = 0.0068 \text{ m}^2$, $\Delta P = 101.325 \text{ kPa}$ and $w = 1.33 \text{ mm}$).

As the geometric parameters and the pressure differential are fixed, the only controllable parameter to avoid the virtual leakage from exceeding the capacity of Turbomolecular Vacuum Pumps is Permeation constant (K). This will serve as a primary **Minimization Objective** in Granta selection phase.

4.8.1 Monolithic Approach

As the monolithic elastomers ideally maintain a perfectly elastic physical seal, the system's baseline pressure is strongly dictated by the inherent porosity of the rubber matrix. Elastomers are generally more permeable than thermoplastics. So, to keep the gas permeation rates within the capacity of Turbomolecular pumps, the base polymer must be optimized for molecular density.

4.8.2 Composite Approach

The thermoplastic (PTFE or FEP) jackets possess exceptional barrier properties. The highly crystalline molecular structure (in case of high crystallinity) results in a significantly lower Permeation Coefficient (K) in comparison to cross-linked counterpart. However, this advantage is nullified in case of dynamic applications. As mentioned in section 4.6.2, resultant permanent deformation will create macroscopic leak paths contaminating the vacuum. Thereby, the material itself will prevent gas diffusion, but the overall component will physically fail to maintain the necessary vacuum interface.

4.9 Outgassing Kinetics & Chamber Contamination

During the required 150°C bake-out step, the unreacted elements within the bulk matrix and the absorbed species (moisture and gases) vaporize into the chamber. If these outgassed molecules are Volatile Condensable Materials (VCM), they will possibly react and contaminate the electromagnetic lenses, permanently degrading the microscope's optical resolution.

The rate and total mass of the desorption are strictly governed by the aerospace testing standards. Under *NASA Outgassing Database* and *ASTM E1559 (Standard Test Method for Outgassing Kinetics)* [10], The total Outgassing Rate (OGR) is calculated using a Quartz Crystal microbalance to measure the mass deposition $\dot{m}_q$ over the sample area A_s factored by the geometric view factor (F_q).

$$OGR(T_s, t) = \frac{F_q \cdot \dot{m}_q(T_s, T_q, t)}{A_s}$$

Furthermore, the overall mass degradation of the seal is calculated as Total Mass Loss (TML).

$$TML(T_s, t) = 100 \cdot \left[\frac{F_q \cdot \dot{m}_q(T_s, T_q, t)}{m_{s+h}(i) - m_h} \right]$$

Thereby, to guarantee the chamber contamination free and keeping the lens safe, the target material must pass the certification limits defined by *ASTM E595 (Standard Test Method for TML and CVCMS from Outgassing in a Vacuum Environment)*.

So, during the material selection phase in Granta, the search parameters will enforce the following hard constraints:

- **Total Mass Loss (TML):** 1.0%
- **Collected Volatile Condensable Material (CVCMS):** 0.1%

4.9.1 Architecture Considerations

Monolithic O-rings that are generally manufactured from standard commercial rubbers are usually packed with plasticizers and unreacted monomers. During bake-out cycles, these characteristics pose a risk of vaporization and boiling of several contaminating elements within the chamber and destroy the SEM optics. Therefore, it is required for the monolithic design to undergo aerospace-grade post cured formulation.

On the other hand, the composite rings with PTFE coating are incredibly pure and ideally possess zero outgassing, rendering them a perfect fit for this parameter.

4.10 Chemical Resistance

While the steady state environment of the SEM is effectively an inert vacuum (after regular N₂ purging during each cycle), the dynamic seal is exposed to harsh chemical solvents during the loading and preventive maintenance stages. Standard cleaning protocols for vacuum applications dictate that the system (specimen stage, flange and surrounding hardware) should be wiped with organic solvents, usually Isopropyl Alcohol (IPA), Acetone or Ethanol, to remove human skin oils (during handling) and particulate contaminants.

If the selected elastomer is chemically incompatible with these cleaning agents, it could result in solvent absorption by the matrix resulting in volumetric swelling. According to Parker O-ring Handbook, the material is degraded in two ways.

First, the solvent chemically extracts the internal plasticizers through (through like-dissolve-like phenomenon) resulting in the material becoming brittle and prone to micro-cracking once the solvent evaporates. Second, when the swollen seal is closed after loading and the chamber experiences the pressure gradient (followed by a subsequent thermal bake-out), the trapped liquid solvent vaporizes violently. This rapid expansion, normally termed as explosive decompression causes the material to physically blister and rupture and releasing massive amounts of hydrocarbons in the chamber that can contaminate the chamber and affect the optics.

To survive these standard cleaning procedures, the target elastomer must possess a stable, chemically inert structure. As the standard databases usually quantify this parameter qualitatively rather than through a numerical calculation, the material selection through Granta will utilize the bounded index ratings. To pass the screening phase, the candidate material should possess a minimum rating of Good to Excellent for Resistance to Alcohols and Organic Solvents.

4.10.1 Monolithic Approach

As the monolithic design lacks the protective thermoplastic jacket, the base polymer matrix should be compatible with the direct solvent exposure. Many industrial grade elastomers (based on Nitriles and Silicon) undergo extreme swelling when exposed to cleaners like acetone. So, highly stable, fluorinated formulations are advised for this architecture to possess the necessary chemical inertness to survive without a jacket.

4.10.2 Composite Approach

The primary justification for encapsulated O-rings is their chemical resistance. The thermoplastic jackets are universally inert to standard laboratory solvents, completely preventing volumetric swelling and protecting the inner elastomeric core.

4.11 Final Design Topology Selection

As explained in the preceding parametric analysis, the selection between an encapsulated and the monolithic topology depends on the fundamental engineering trade-off between chemical inertness and mechanical elasticity.

According to sole material characteristics, the encapsulated design because of its PTFE/FEP protective coating is superior as a protective barrier. This was noted down in sections 4.8, 4.9 and 4.10 that the un-crosslinked thermoplastic jacket provides an ultimate shield against solvent swelling, explosive decompression and outgassing contamination ($TML \leq 1.0\%$).

However, these chemical advantages are rendered functionally obsolete by the dynamic mechanical requirements of the SEM door during operation. A vacuum seal must physically conform to the flange geometry and establish proper vacuum sealing first before its chemical inertness can be utilized.

From section 4.6, we verified that the rigid Shore D thermoplastic jacket lacks the micro-scale conformability to seal properly against the flange asperities without warping the door hinges. Most critically, we mathematically proved in section 4.7 that the encapsulated design fails the dynamic elastic recovery constraint. Upon the application of 150°C thermal bake-out and 25% mechanical compression, the plastic jacket inevitably suffers from cold flow (creep). Upon opening the chamber door, the yielded jacket fails to recover to acceptable limits, creating permanent macroscopic wrinkles that destroy the vacuum interface. This parametric conclusion aligns directly with Barnwell-O-ring material selection overview [7], whose technical guidelines advise against the use of PTFE-encapsulated O-rings or dynamic operations due to rapid jacket fatigue.

Because the mechanical spring recovery is the absolute pre-requisite for the dynamic face seal, the encapsulated topology is officially rejected at this stage. To ensure that the seal survives the cyclic clamping and thermal expansion without yielding, the system must proceed with the monolithic topology. This baseline will be transferred to Granta EduPack software, that would utilize the derived performance indices and design constraints to isolate a highly specialized elastomeric formulation that would inherently provide mechanical conformability, chemical inertness and low outgassing rates simultaneously.

5. Function, Constraints & Objectives

With all the main parameters and architecture selected, we can define the main function, constraints and maximizing objectives for this analysis.

- ❖ The O-ring must maintain a dynamic, hermetic seal that isolates the SEM's vacuum chamber from atmospheric air during mechanical motion. To ensure system vacuum integrity across multiple cycles, the component must exhibit ultra-low outgassing, high elastic recovery, and structural resistance to critical temperature and pressure conditions.

As for the constraints, we summarized them in the Table 2.

Table 2. Constraints & Considerations

Property	Derived Value/Limit	Granta Input	Overview	References/Standards
Max. Service Temperature	>150°C ($\Delta T_{\text{calc}} = 130^\circ\text{C}$)	Maximum Service Temperature (Limit Stage)	Ensuring the polymer selected does not melt or degrade during chamber bake-out procedure	JEOL Installation Requirements, ASTM E595
Compressive Yield Strength	>4.6 MPa	Minimum Yield strength/elastic limit (Limit Stage)	Prevent permanent deformation under ΔP during steady state	ISO 3601-2 (25% compression set), JEOL SEM database
Thermal Expansion	$\left(\frac{\Delta V}{V}\right) < 10\%$ (direct relationship to α) or MINIMIZE	Thermal Expansion Coefficient (Limit Stage)	Prevents volumetric “gland overfill” and extrusion during 150°C bake-out.	Parker O-ring Handbook
Surface Hardness	70-85 Shore A	Hardness Shore A (Limit Stage)	To fill microscopic asperities and avoid vacuum extrusion	Parker O-ring Handbook, ASTM D2240
Compression Set	$\leq 20\%$	Compression Set (Limit Stage)	Guarantee the seal elastically recovers its thickness to prevent virtual leaks.	SAE AMS 3216, ASTM D395
Gas Permeability (K)	MINIMIZE	(put proxy in Limit stage-permeability to gases) and post confirmation with data bases	To reduce the steady state influx of the moisture and other gases from the laboratory environment during cycles.	Fick’s Law of Diffusion, Henry’s Law
Outgassing (TML and CVCM)	TML $\leq 1.0\%$, CVCM $\leq 0.1\%$	MINIMIZE (put proxy-water absorption in 24 hours) and post confirmation with NASA databases	Restricts total mass of vaporized material to protect the lens optics.	ASTM E1559, ASTM E595
Chemical Resistance	Excellent/Good	Resistance to Alcohols & Organic	Prevents chemical swelling and blistering when	High Vacuum-Cleaning Protocols,

		Solvents (Limit Stage)	exposed to lab cleaning solvents.	Parker O-ring Handbook
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6. Material Selection via GRANTA

6.1 Performance Indexes

6.1.1 M_1 (Mechanical Conformability – Maximum Elastic Deformation)

The primary mechanical function of the seal is to prevent the outside atmospheric gas and contaminants from bypassing the chamber-door interface. To accomplish this, the intended elastomer must physically deform to fill up the microscopic asperities of the metallic flange. The driving force for this deformation is the atmospheric clamping force (F_c) acting on the external surface area (exposed to outside laboratory setting) of the SEM door (A_{door}) due to the operational pressure differential (ΔP).

$$F = \Delta P \cdot A_{door}$$

This macroscopic clamping force is transferred completely to the O-ring, defining a compressive contact stress ($\sigma_c = \frac{F}{A_c}$). As for safe and normal operation, we require the elastomer to operate within the linear elastic regime (at maximum, exhibiting plastic deformation within the allowable compression limits defined earlier). So, within this regime, the relationship between the applied stress and resultant deflection (strain) is provided by Hooke's Law ($\sigma = \epsilon \cdot E$).

For an O-ring with a cross-sectional diameter (d), the strain is defined as ($\epsilon = \delta/d$). By the substitution of clamping force directly into Hooke's law, we can calculate the physical deflection of the seal.

$$\frac{F}{A_c} = E \cdot \left(\frac{\delta}{d} \right)$$

Isolating the deflection (δ) from the system reveals how the resultant forces interact with the material's stiffness:

$$\delta = \frac{F \cdot d}{A_c \cdot E}$$

$$\delta = \left[\frac{\Delta P \cdot A_{door}}{A_c} \right] [d] \left[\frac{1}{E} \right]$$

Although the lower Young Modulus will mathematically increase the overall deflection (δ), the O-ring cannot be infinitely soft to deform and cover off the asperities. If the operational contact stress exceeds beyond the material's compressive yield strength (σ_{yc}), the polymeric chains slip beyond the elastic recovery point and settle at minimum free energy state (plastic deformation). Thereby, upon loading/unloading, a yielded seal will fail to elastically recover and create virtual leakage that can contaminate the chamber.

So, the maximum allowable contact stress before operational failure can be defined as:

$$\frac{F}{A_c} \leq \sigma_{yc}$$

So, to find the maximum elastic deformation the seal can achieve safely before yielding, we can substitute the yield strength (σ_{yc}) back into the deflection equation instead of the applied stress like:

$$\delta_{max} = d \times \left(\frac{\sigma_{yc}}{E} \right)$$

So, in accordance with Ashby's methodology, the performance equation is now strictly governed by a fixed geometric constant ($d = 5.33mm$) and a variable material index. To ensure that the seal can maximize its deflection (within the allowable constrained limits) to conform to the asperities without significant plastic deformation, the material parameter can be optimized.

So, we can define our first performance index (M_1) to be **maximized** as:

$$M_1 = \left(\frac{\sigma_{yc}}{E} \right)$$

Granta Execution: A standard log-log Ashby chart will plot the Compressive Yield Strength (σ_{yc}) on the Y-axis against the Young Modulus (E) on the X-axis. To maximize the index (M_1), a selection guideline with a slope of +1 will be generated, defined by the performance equation.

$$\log(\sigma_{yc}) = \log(E) + C$$

6.1.2 M_2 (Thermo-Mechanical Stability)

During the maintenance cycle, the SEM chamber and the door flange are subjected to the already concerned high vacuum bake-out at 150°C to desorb the moisture. When the elastomer is subjected to this temperature gradient (ΔT), the polymer matrix tries to expand. The theoretical thermal strain (ϵ_{th}) of the material is provided by the product of the temperature gradient and coefficient of thermal expansion (α).

$$\epsilon_{th} = \alpha \cdot \Delta T$$

In the operational setting, the O-ring is not allowed to expand freely; it is heavily confined within the rigid, machined geometry of the metallic door flange (dictated by reference *AS568-377* compatible geometry). As the thermal expansion of the steel (or alloyed) flange is negligible in comparison to the elastomer, the groove acts as a rigid body constraint. This means that the elastomer's attempt to expand is completely resisted by the groove setting on one side and door flange on the other side.

According to Hooke's law, this restrictive strain is converted to internal compressive thermal stress (σ_{th}) within the polymer matrix.

$$\sigma_{th} = \epsilon_{th} \cdot E$$

Substituting ϵ_{th} in the above equation, we have:

$$\sigma_{th} = E \cdot \alpha \cdot \Delta T$$

As the temperature rises, the internal thermal stresses also rise. If this induced stress exceeds beyond the Compressive Yield strength (σ_{yc}) of the material, the polymer chains will be crushed against the walls of the flange and undergo permanent deformation (beyond allowable constrained limits).

When the SEM cools back down to the room temperature at the end of the cycle, the yielded O-ring will not elastically shrink back to the desirable intended geometry and size, instantly causing a massive vacuum leak. So, the absolute limit to survive this thermal gradient during normal operational cycle without yielding is:

$$\sigma_{th} \leq \sigma_{yc}$$

By substituting the thermal stress equation back into the above explained limit (failure constraint) equation, we link the thermal limits directly to the mechanical limits of the elastomer:

$$E \cdot \alpha \cdot \Delta T \leq \sigma_{yc}$$

Rearranging the inequality to isolate the thermal gradient (operational thermal range) will provide the maximum thermal shock the seal will survive before failure like:

$$\Delta T \leq \left[\frac{\sigma_{yc}}{E \cdot \alpha} \right]$$

To guarantee that the seal survives the required 150°C bake-out temperature without suffering more than allowable thermally induced compression set, the material should be capable of withstanding the largest possible operational temperature differential (ΔT).

Therefore, we require the secondary performance index (M_2) to be maximized as:

$$M_2 = \left(\frac{\sigma_{yc}}{E \cdot \alpha} \right)$$

Granta Execution: A custom user-defined log-log chart will map the Conformability ratio ($\frac{\sigma_c}{E}$) on the Y-axis against the coefficient of Thermal expansion (α) on the X-axis. To maximize the M_2 , a selection guideline with slope +1 will be translated, extending towards the top-left quadrant to isolate materials with maximum elasticity and minimum thermal expansion. This is defined by the equation.

$$\log\left(\frac{\sigma_c}{E}\right) = \log(\alpha) + C$$

6.1.3 M_3 (Permeability-Compensated Conformability)

As already discussed, even under full compression, the elastomeric polymers are inherently porous. The transportation of atmospheric moisture into the vacuum chamber through the seal is governed by Fick's law of diffusion. The volumetric leak rate (Q) is dependent on the material's Gases Transmission Rate (P_{gas}), the exposed surface area (A), the pressure gradient (ΔP) and physical barrier thickness (w). Mathematically, it can be expressed as:

$$Q = \frac{(P_{gas}) \cdot A \cdot \Delta P}{w}$$

(This equation is derived from the permeation coefficient equation but made adaptable to available Granta parameters).

Standard Ashby material selection indices for compressive gas barriers (*Ansys-Performance Indices Handbook*) [11] are derived under the assumption of a “thickness-free” structural panel. In such macroscopic panels, if the wall is leaking too much gas, the engineer can increase the thickness (w) arbitrarily to reduce the leak rate (Q).

However, a dynamic SEM operates within a fixed geometrical constraint (as for our case, defined entirely by the machined dimensions of AS568-377 flange groove. As the overall cross-sectional thickness cannot be increased freely without preventing the door from closing, these indexes are mathematically invalid for our O-ring selection.

So, instead of having a free thickness, the O-ring is subjected to “Compressive thinning”. To achieve mechanical sealing dictated by the first performance index (M_1), the material must be compliant (a lower Young’s Modulus).

However, as the atmospheric force crushes the highly compliant O-ring, its physical barrier thickness is drastically compressed (w_c). Looking back at Fick’s law equation, a smaller w_c in the denominator will exponentially increase the leak rate (Q).

This generates an engineering paradox; the material must be soft to conform according to the flange geometry and fill up the asperities ($\frac{\sigma_c}{E}$). But the more the material deforms, the thinner the O-ring gets, allowing for vapours and gases to pass through the interface and enter the chamber.

As the system cannot solely rely on physical thickness to stop the leak, the material itself must possess a dense molecular structure to compensate for this compressive thinning. To find the optimal balance, this mechanical conformability objective must be coupled with vapor permeability penalty.

So, the third index can be defined and required to maximize as:

$$M_3 = \left(\frac{\sigma_{yc}}{E \cdot P_{gas}} \right)$$

Granta Execution: A custom log-log chart will map the Conformability ratio ($\frac{\sigma_c}{E}$) on the Y-axis against the Water Vapor Transmission rate (VTR) on the X-axis. To maximize M_3 , a selection guideline with a slope of +1 will be translated towards the top-left quadrant to identify materials that support maximum mechanical elasticity and spring back while acting as a dense molecular barrier at the same time. We can utilize the following equation:

$$\log \left(\frac{\sigma_c}{E} \right) = \log(P_{gas}) + C$$

Table 3. Performance Indexes for Granta Selection Phase

Index	Design Purpose	Performance Index (to Maximize)	Ashby Log-log Line
M_1 (Mechanical Conformability)	Maximize elastic deformation into		

	the flange without suffering plastic yield	$M_1 = \left(\frac{\sigma_{yc}}{E} \right)$	$\log(\sigma_{yc}) = \log(E) + C$
M ₂ (Thermomechanical Stability)	Minimize thermal expansion to prevent heat-induced compression set during bake-out	$M_2 = \left(\frac{\sigma_{yc}}{E \cdot \alpha} \right)$	$\log\left(\frac{\sigma_c}{E}\right) = \log(\alpha) + C$
M ₃ (Permeability compensated conformability)	Minimize gases and vapor permeation while maintaining mechanical elasticity.	$M_3 = \left(\frac{\sigma_{yc}}{E \cdot P_{gas}} \right)$	$\log\left(\frac{\sigma_c}{E}\right) = \log(P_{gas}) + C$

6.2 Material Selection Methodology (GRANTA)

To identify the optimal elastomeric material for our O-ring that would be installed in the SEM's dynamic HV environment, a systematic filtering protocol was established using Granta EduPack. The methodology was structured in two primary stages: Database Taxonomy Definition and Constraint Application strategy.

6.2.1 Stage 1 (DataBase Taxonomy Definition)

To eliminate the extra material classes from our analysis firsthand (such as metals, ceramics, composites etc), we focused strictly on Polymers from the start. So, the database was restricted from the start at the taxonomy level i.e., our selection procedure exclusively utilized Level 3-Polymers. The search process was filtered explicitly to *Materials Universe* $\rightarrow$ *Polymers*. We did not sub-filtered the database to elastomers only to access a wide database that would then be filtered out selectively from our structured criteria-based filtration.

Filter Results: The database pushed **834** qualified materials out of 1030 materials in the specific Level 3 Material regime.

6.2.2 Stage 2 (Limit Stage Definition)

At this stage, the engineering parameters derived from the constraints section were systematically applied. Since all the constraints were not directly available in the Granta EduPack software, we categorized the constraints into 2 application methods: Boolean Limits (Direct and Proxy variables) and post-selection verification.

6.2.2.1 Boolean Limits (Direct and proxy Variables)

These constraints were employed in the Limit stage in the second stage of Tree filtration for immediate elimination of unqualified polymers.

- **Thermal Operational Limits (Direct):** The temperature ceiling was set in accordance with the constraint limits defined above from the OEM manufacturer database (minimum temperature $<20^{\circ}\text{C}$ and the maximum temperature $>150^{\circ}\text{C}$) to ensure that the material survives the thermal bake-out process.
- **Compressive Yield Strength (Direct):** The ensure that the material would not undergo permanent deformation under normal operation, an engineering safety factor of 1.5 was also incorporated at our already defined limit of $>4.6\text{ MPa}$. So, the absolute limit was defined at 6.6 MPa .
- **Surface Hardness (Direct):** Since the standard material databases define the structural hardness using the conventional Vickers scale, which is invalid for elastomers. Thereby, the constraint limits of Shore A (70-85) hardness were translated to an effective Young's Modulus using (*Gent-Engineering with Rubber: Design of Rubber Components*) [12]. The empirical formula led to the Modulus limits of 3-8 MPa. However, upon detailed inspection in advanced settings, Hardness Shore A was located and applied. The empirically calculated Young's Modulus was kept as a further relative check.
- **Chemical Resistance (Direct):** To prevent solvent swelling and explosive decompression of volatile elements (upon exposure during acetone cleaning or present within the laboratory environment), the durability of Water (fresh) was restricted to "Excellent" and for Organic Solvents, it was restricted to "Excellent and Acceptable".
- **Compressibility Setting (Direct):** To guarantee allowable elastic recovery after each cycle (unloading/loading) of the specimen, essential for normal operation of SEM, the compression setting of $\leq 20\%$ is specified in the limit stage. We chose the temperature to be 100°C for this requirement as it was the closest to our target operational ceiling of 150°C .

6.2.2.2 Post Selection Verification

The specific polymers that will survive the complete selection process would be subjected to external manual verification to confirm the specialized dynamic and trace-chemical properties.

- **Compression Setting (beyond 100°C):** The candidate materials will be cross-referenced against ASTM D395 and SAE AMS 3816 data sheets to confirm the compression set of $\leq 20\%$ to guarantee appropriate elastic recovery after each cycle. This is required as Granta EduPack has its Limit stage having a maximum at 100°C .
- **Optical Outgassing Purity:** The candidate materials will be validated against the NASA Outgassing Data for Selecting Spacecraft Materials (as per ASTM E1559, ASTM E595

standards) to explicitly confirm the Total Mass Loss (TML) $\leq 1.0\%$ and Collected Volatile Condensable Material (CVCM) $\leq 0.1\%$.

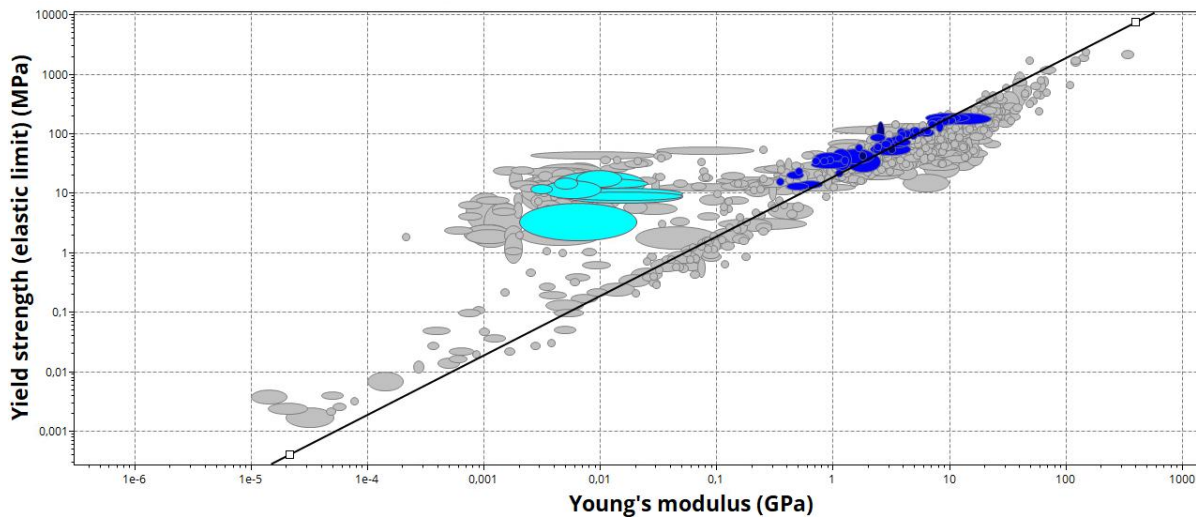
Filter Results: After the application of Boolean Limits, Granta specified **52** qualified materials from Stage 2.

6.2.3 Stage 3 (Graphical Optimization through Performance Indexes)

After the application of absolute boundary constraints, the surviving material pool is evaluated further by subsequent application of Performance indices to categorically remove the materials that do not qualify according to our requirements.

6.2.3.1 Conformability Index (M_1)

As already defined in section 5.1.1, the logarithmic Ashby chart is generated, plotting Yield Strength (Elastic Limit) on the Y-axis against Young's Modulus (E) on X-axis. To **maximize** the performance index, a selection guideline with a slope of +1 was applied to the plot, and the generated line was translated to the top-left quadrant i.e., the region representing maximum compressive resilience with maximum elastic compliance.



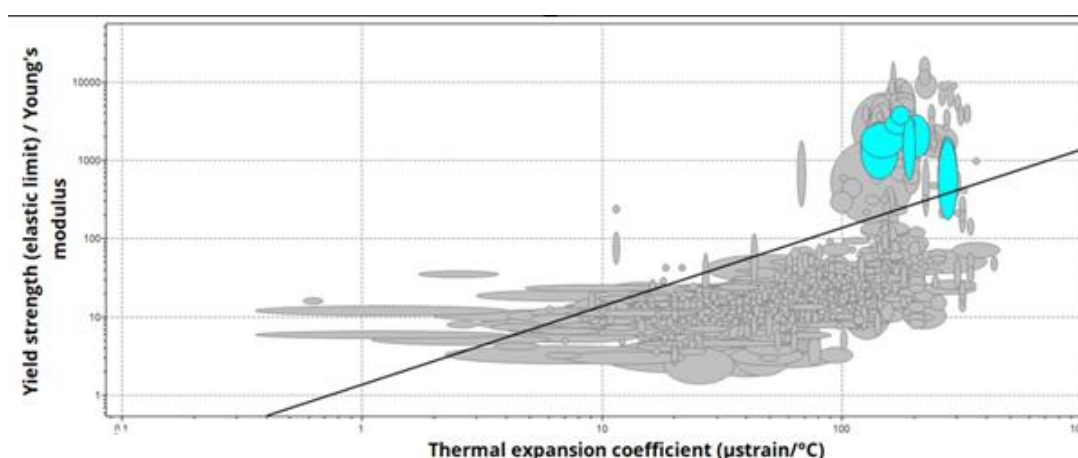
Graph 1. M_1 maximization results (Yield strength (Elastic limit) Vs Young's Modulus)

Filter Results: The graphical filter eliminated the stiffer lower-yielding rubbers, isolating the polymers depicting maximization of M_1 performance index. This filter narrowed down the candidate materials list to **59** entities.

6.2.3.2 Thermo-Mechanical Stability Index (M_2)

Remaining polymers are now subsequently evaluated for their resistance to volumetric expansion during the 150°C thermal bake-out. This is carried out by formulating a second logarithmic chart to map out the thermo-mechanical stability index (M_2).

The Y-axis was defined by the previously derived Conformability Index (M_1), and the X-axis was assigned to the Thermal Expansion Coefficient (α). A selection guideline with a slope of +1 was applied and translated to the top-left quadrant. This vector **maximizes** the performance ratio, isolating the elastomers that retain exceptional dynamic spring recovery while undergoing minimum dimensional swelling under extreme thermal gradients.



Graph 2. M_2 maximization results (M_1 Vs Thermal Expansion Coefficient)

Filter Results: The intersection of absolute boundary limits between the M_1 mechanically focused plot and M_2 thermally focused plot successfully eliminated all the conventional rubbers. This cross-referenced approach narrowed down the candidate list to just **9** materials.

6.2.3.3 Permeability Compensated Conformability Index (M_3)

The final performance index evaluated the simultaneous requirements of mechanical compliance along with vacuum integrity. In the theoretical derivation of volumetric leak rate (Q), the compressed seal thickness (t) is strictly constrained by the physical geometry of the O-ring gland. However, the Granta's *Native Performance Index Finder* inherently models the thickness as a “free variable” when calculating the gas/vapor barriers as the equations are hard-coded and immutable.

As these parameters cannot be user modified to reflect the fixed geometries, so direct input of M_3 was not possible. To navigate through the software's limitation while preserving the underlying physics, the index was formulated as a functional trade-off utilizing the native “Gas/vapor barrier in compression” model.

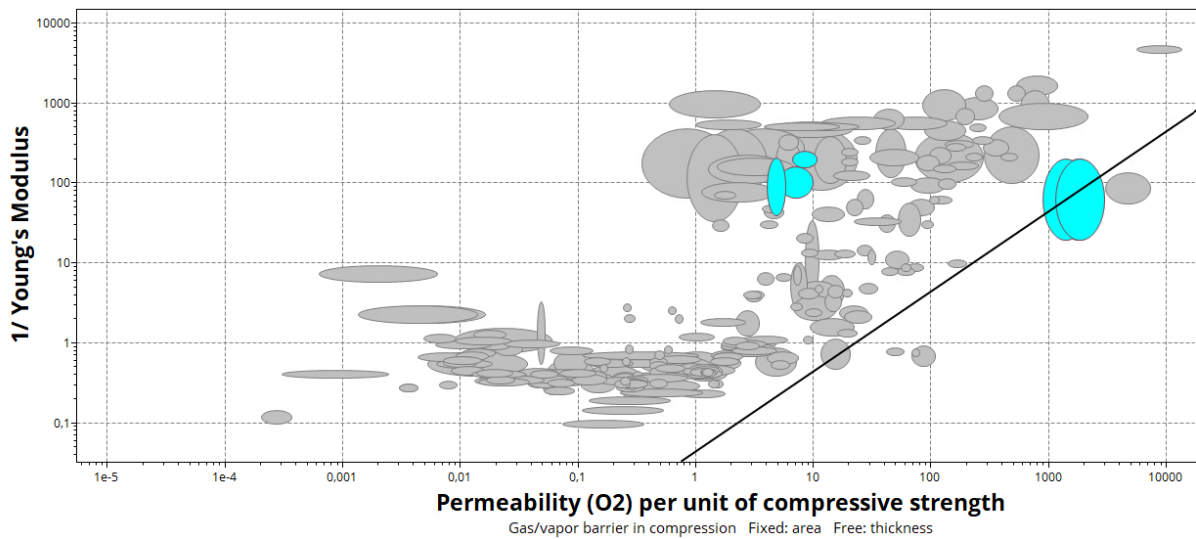
According to this hardcoded physical model, minimizing the permeability for compression loaded barrier forces us to modify the index equation to acquire $\left(\frac{P_{gas}}{\sigma_{yc}}\right)$ as a parameter. To integrate this with the previously established requirement and the derived performance index, the following Pareto front was generated:

- The native barrier index $\left(\frac{P_{gas}}{\sigma_{yc}}\right)$ was kept on X-axis (to minimize) and the purely elastic compliance factor $\left(\frac{1}{E}\right)$ was plotted on the Y-axis (to maximize). By applying the selection guideline with a slope of +1 and translating it towards the top-left quadrant, the graphical filter optimized the ratio of the two axes. So, while bypassing the requirements of software, the M_3 is now modified as (M_3^*):

$$M_3^* = \frac{\left(\frac{1}{E}\right)}{\left(\frac{P_{gas}}{\sigma_{yc}}\right)}$$

And the consequent Ashby logarithmic equation is modified as:

$$\log\left(\frac{1}{E}\right) = \log\left(\frac{P_{gas}}{\sigma_{yc}}\right) + C$$



Graph 3. Modified M_3 (M_3^*) maximization results $\left(\frac{1}{E} \text{ Vs } \frac{P_{gas}}{\sigma_{yc}}\right)$

Filter Results: The final index further refined the target materials for our application to **6** finalists. These materials will now be subjected to *NASA database*, *ASTM E1559* and *ASTM E595* to be verified for the remaining constraints that were not accounted for within Granta EduPack database.

The 6 finalist materials after our structured selection process are.

- Fluoro elastomers (FEP/AFLAS)
- Fluoro elastomers (FEP/AFLAS, 15-30% Carbon Black)
- Fluoro elastomers (FKM, 20-35% Carbon Black)
- Fluoro Silicone (FVMQ, heat-cured, 10-30% fumed Silica)
- Silicone (VMQ, heat-cured, 10-30% fumed Silica)
- Silicone, phenyl type (PVMQ, heat-cured, 10-30% fumed Silica)

6.3 Post Selection Verification Phase

These materials will now be referenced against *NASA databases* (to checking TML, CVCM is within proposed allowable limits) and *SAE AMS 3216, ASTM D395 (for compression setting)* as already specified in our post-selection procedure.

We will check the materials for both the outgassing and compression setting. If the material's data is not available under compression setting data sheets i.e., ASTM D395 and SAE AMS, we will consider the material "Approved" as it was able to keep its compression setting till 100°C (this check is added as an extra verification procedure) and for having a safe estimate, will verify the Maximum Operating temperature for the material and verify if its above 150°C. However, in case of outgassing, if the material is not present in NASA database and ASTM E1559 i.e., the material is not certified, and we will "Reject" it as it is a critical parameter for the safe operation of SEM.

The information regarding the materials and their post-selection results can be presented in a tabular form.

Table 4. Material Selection Phase results (Granta and Post-verification)

Materials/Properties	FEP/AFLAS	FEP/AFLAS, 15-30% Carbon Black	FKM, 20-35% Carbon Black	FVMQ, heat cured, 10-30% fumed Silica	VMQ, heat cured, 10-30% fumed Silica	PVMQ, heat cured, 10-30% fumed Silica
Yield Strength [σ_{yc}] (MPa)	12 - 18	13 – 24	12.8 – 17.6	8.2 – 12.1	7 – 11.5	8 – 10.5
Elastic Modulus [E] (MPa)	4 - 6.23	6.23 – 15	4.9 – 25	5 – 50	5 – 50	5 – 50
Hardness [Shore A]	70	70 – 83	65 – 93	70	75	75
Durability-Water (fresh)	Excellent	Excellent	Excellent	Excellent	Excellent	Excellent
Durability-Organic Solvents	Excellent	Excellent	Excellent	Acceptable	Acceptable	Acceptable
Min and Max Service Temperature [ΔT] (°C)	-8 - 230	-13 - 230	-19 – 220	-57 - 250	-60 - 250	-90 - 210
Compression set (till 100°C) (%)	15-20	15 - 20	9 – 25	5 – 10	5 - 10	5 - 10

Post-Selection Verification						
TML (%)	Uncertified	Uncertified	0.15	0.07	0.04	0.17
CVCM (%)	Uncertified	Uncertified	0.01	0.03	0.00	0.03
Compression Set (Beyond 100°C)	30 - 35	30 - 35	10 - 15	12 - 18	10 - 15	15 – 30 (cure dependent)
RESULT	FAIL	FAIL	PASS	PASS	PASS	PASS

6.4 Weighted Decision Matrix

As from the post-selection verification phase, we finalized with four elastomeric families (FKM, FVMQ, VMQ, PVMQ) that satisfy the thermo-mechanical boundary conditions required for our operation. However, a final ranking is required to determine the best material for our application, based on which one maximizes successful operational cycles under our defined constraints. Because all the remaining four candidates exceeded the structural requirements (Modulus, Compressive Yield Strength and Thermal Ceiling) by significant margins, these parameters can be classified as “baseline parameters” as they cannot further filter out the selection and thereby removed from further relative scoring.

So, we moved with the Weighted Decision Matrix evaluation procedure to map out the materials against the critical operational and environmental vulnerabilities of a high-vacuum SEM seal. All these parameters are already validated and all the families cleared them. However, to find out the best among all, these parameters are reconsidered based on their importance and criticality. Each candidate will be scored on a 1-10 scale across the five operational parameters.

- **Organic Solvents Resistance (Weight: 25%):** Highly Critical. The seal must survive routine physical wipe-downs with acetone and iso-propyl alcohol without volumetric swelling, virtual leaking or chemical reaction with the agent.
- **NASA TML (Weight: 25%):** Highly Critical. Minimizing the total mass loss directly governs the ultimate baseline vacuum achievable by the SEM pumps.
- **NASA CVCM (Weight: 25%):** Highly Critical. Minimizing the condensable material introduction within the chamber is the only way to ensure that the optics remain free of carbon contamination.
- **Compression Setting (above 100°C) (Weight: 15%):** Critical. It dictates the dynamic elastic recovery and physical lifespan of the O-ring after repeated thermal bake-out cycles. This has been given less weight as the materials were subjected to a similar parameter (both under 100°C within Granta limitation and above 100°C from ASTM standards).

- **Water Resistance (Weight: 10%):** Secondary. Although the climate of the laboratory is strictly monitored, atmospheric moisture resistance prevents the localized hydrolysis over years of operation. Even though it's an important parameter, it was weighted lower as all the materials scored "Excellent" in this regime.

So, the weighted matrix can be formulated as:

Table 5. Weighted Matrix for final selection

Evaluation Criteria	Weightage	FKM, 20-35% Carbon Black	FVMQ, heat cured, 10-30% fumed Silica	VMQ, heat cured, 10-30% fumed Silica	PVMQ, heat cured, 10-30% fumed Silica
Organic Solvent Resistance	0.25	10	8	3	3
NASA TML (≤ 1.0)	0.25	8	9	10	9
NASA CVCN ($\leq \%$)	0.25	9	8	10	9
Compression Setting (above 100°C)	0.15	10	8	9	6
Water Durability	0.1	10	10	10	10
Final Weighted Score (/10)	1.00	9.25	8.45	8.1	7.15

As all the final candidates exhibited acceptable baseline outgassing and thermal recovery profiles, the definitive discriminatory parameter was the operational resilience against the organic solvents. In case of high-vacuum optical systems, routine physical maintenance dictates that the dynamic seals are repeatedly wiped with aggressive cleaning agents (e.g., Isopropyl Alcohol, Acetone and Ethanol etc) to prevent the contamination of electron beam.

Looking at the molecular structure of standard silicone architectures (VMQ, FVMQ, PVMQ, the polymer's chemical compatibility reveals a fatal operational risk. The siloxane backbone (Si-O) possesses very high molecular free volume. When exposed to standard laboratory solvents, the silicone matrix rapidly absorbs the fluids, resulting in extreme volumetric swelling. In a dynamic SEM flange, using these materials would result in dimensional extrusion from the flange groove, creating virtual leaks and causing outgassing of absorbed solvents during pump-downs.

On the other hand, the dense fluorinated network of FKM (Fluoro-elastomer) exhibited superior chemical inertness. The highly electronegative fluorine atoms shield the carbon backbone, creating a low free-volume structure that prevents solvent diffusion. This has been verified within industry standard fluid compatibility guidelines (*Parker O-ring Handbook, ORD 5700*), FKM exhibits an excellent compatibility rating for organic laboratory solvents, resisting absorption and maintaining dimensional stability under routine maintenance.

Thereby, **FKM (Post cured by Carbon Black 20-35%)** achieves the highest overall operational score and is selected as the sole, technically viable solution for the dynamic SEM seal, fulfilling all the Granta and NASA, ASTM criteria.

7. Sustainability Analysis (Eco-Audit)

To evaluate the environmental and economical footprint of our O-ring, a complete Lifecycle Assessment was modelled using *Granta EduPack Eco Audit Tool*. The audit tool had multiple sections associated with the material itself, the manufacturing process, transportation, and its usage. We will explain the parametric input one by one in the following headings.

7.1 Material, Manufacture & End of Life Section

This section required input for the following parameters:

- **Quantity:** (1). Although, JEOL produces several thousand electron optic instruments per year worldwide, but for component level eco-audit, it's essential to evaluate the environmental impact on a "per machine/product" basis so that it can be easily scaled by the manufacturer later depending upon its production scale.
- **Material:** The final material from our selection procedure i.e., FKM (Carbon black 20-35%) is our material of interest.
- **Recycled content:** (Virgin 0%). AS FKM is a thermoset and not a thermoplastic, the polymer chains undergo permanent chemical crosslinking during vulcanization procedure and because of these permanent bonds, the material cannot be melted and reformed. So, 100% of the O-ring must come from the virgin fluorocarbon.
- **Mass Calculation:** For the mass calculation, we must refer to section 4.2 to get the geometrical parameters required to calculate the volume and ultimately mass of the O-ring.

So, Inner Diameter (ϕ_{IN}) = 253.57 mm and Cross-section (ϕ_{CS}) = 5.33 mm.

Mean diameter (ϕ_M) = 258.9 mm

$$\text{As, } V = \frac{\pi^2}{4} \times \phi_M \times (\phi_{CS})^2$$

$$V = \frac{\pi^2}{4} \times (258.9\text{mm}) \times (5.33\text{mm})^2 = 18.147\text{cm}^2$$

From Granta, we can get $\rho_{FKM} = 1.85 \text{ g/cm}^3$

$$\text{So, } m = \rho \cdot V$$

$$m = \left(1.85 \frac{g}{cm^3}\right) (18.147 cm^2) = 33.573 g = 0.0336 kg$$

- **Primary Manufacturing Process:** (Polymer Molding-Injection). According to GRANTA database, the best manufacturing technique adopted for FKM based materials is Injection molding.
- **End of Life:** (Landfill). As FKM is heavily fluorinated, it should not be incinerated under standard recycling conditions because its combustion could release highly toxic and corrosive Hydrogen Fluoride (HF) gas which is a massive environment hazard. Also, it's a thermoset and thereby cannot be recycled. So, inert landfill is the safest option for fluoro-elastomers.

Material, manufacture and end of life ⓘ						
How do I use my own materials or processes?						
Qty.	Component name	Material	Recycled content	Mass (kg)	Primary process	End of life
1	SEM HV O-ring	Fluoro elastomer (FKM,...	Virgin (0%)	0,03376	Polymer molding (injection, blow) ▾	Landfill

Figure 5. Parameters uploaded for Eco-audit (Material, manufacture and EOL)

7.2 Transport Stage

For this section, the complete journey is divided into two parts.

First is the long “intercontinental flight” comprising of around 9000 km. The means of transport was specified to Aircraft, long haul belly freight. We do not need cooling or refrigeration as FKM is stable at room temperature. Another alternative to this is to do “International shipping” through a bulk carrier.

The second stage refers to “regional distribution” within the country after the shipment reaches the airport. This can be carried out by 16-32t Trucks with EURO 5 compliant fuel standard (mandatory within many EU countries) at an expected journey distance of 500 km. An alternative to this transport route can also be provided by opting for “Train-electricity or diesel” depending upon the maximum amount of energy utilized by them. We will move forward with the case study of “SEM in PoliTO’s laboratory” so that both transport routes would be viable.

The optimum means of transportation for our case can be planned out based on the amount of energy utilized by each transport mean (MJ) and the amount of CO₂ equivalent released in the atmosphere (kg).

Table 6. Optimization for Means of Transport

	Energy Consumption (MJ)	Emissions (kg CO ₂ (equivalent))
Stage 1 (9000 km)		
Intercontinental Flight (Aircraft, long haul, belly freight)	3.7	0.26

International Shipping (Sea, bulk carrier)	0.028	0.0022
Stage 2 (500 km)		
Regional Transport (16-32t, EURO 5 truck)	0.045	0.003
Regional Transport (Train-Electricity)	0.01	0.00062
Regional Transport (Train-Diesel)	0.011	0.00079

So, looking at the data for both of our target parameters, we can clearly go for the “International Shipping” by a Sea, bulk carrier for the Stage 1 of our journey and for the Stage 2, we will select the Train-electricity option as it provided the least expenditure of energy and carbon emissions.

Transport ?		
Name	Transport type	Distance (km)
International Shipping	Sea, bulk carrier	9000
Regional Transportation	Train, electricity	500

Figure 6. Parameters uploaded for Eco-audit (Transport)

7.3 Use Stage

- **Product Life:** (5 years). As the FKM will not melt or chemically degrade for decades, the O-ring will be rendered as a maintenance wear item. Standard high-vacuum preventative maintenance (PM) schedules dictate replacing the dynamic seals every 3 to 5 years to guarantee safe operation with zero leaks.
- **Country of Usage:** (Italy). The selected country dictates the Carbon intensity of the electrical grid (e.g. France is heavily nuclear/low carbon dependent; Italy has different mix of conventional and green methods). So, we assume a primary laboratory installation within PoliTO that provides a localized grid baseline for the LCA.
- **Static and Mobile mode:** The SEM itself is a static machine that consumes heavy electricity. However, our functional unit, i.e., O-ring does not use any energy for its operation. So, the energy footprint can be mapped as 0 Joules.

Use ?

Product life: Years

Country of use:

Static mode

☐ Product uses the following energy:

Energy input and output:

Power rating: W

Usage: days per year

Usage: hours per day

Mobile mode

☐ Product is part of or carried in a vehicle:

Fuel and mobility type:

Usage: days per year

Distance: km per day

Figure 7. Parameters uploaded for Eco-audit (Use)

7.4 Results of Eco-Audit

By supplying Granta EduPack *Eco-audit Tool* with the already discussed parameters, we obtained a comprehensive report on the environmental footprint of the chosen material. The complete lifecycle of the component (O-ring) was divided into 5 modules (Material, manufacture, transport, use and EOL) with each module's contribution noted in the overall process' energy expenditure and CO₂ emission.

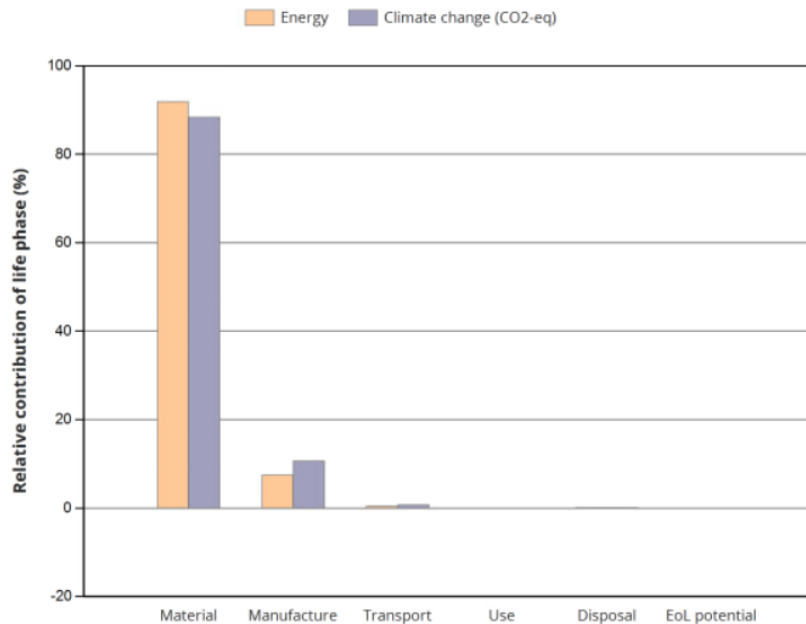


Figure 8. LCA analysis for the O-ring (FKM)

Based on this graph, the discrete numerical values obtained can be organized by focusing first on the absolute and relative primary energy demands, followed by the corresponding greenhouse emissions.

Table 7. Lifecycle stages' contribution mapped

Phase	Energy (MJ)	Energy (%)	Climate change (CO ₂ -eq) (kg)	Climate change (CO ₂ -eq) (%)
Material	7	91,9	0,309	88,4
Manufacture	0,569	7,5	0,0373	10,7
Transport	0,0388	0,5	0,00279	0,8
Use	0	0,0	0	0,0
Disposal	0,00675	0,1	0,000473	0,1
Total (for first life)	7,61	100	0,349	100
End of life potential	0		0	

We can clearly observe from these results that the relative contribution from both the Material and Manufacturing stages comes to almost 99% of the environmental burden, with the Material's stage being our largest contributor for both the energy expense and emissions.

7.4.1 Material Phase Burden

The synthesis of the raw fluoro-elastomer is the single largest contributor to the life cycle, requiring 7.0 MJ of primary energy and generating 0.31 kg CO₂ (eq). On the molecular level, FKM requires an immense energy input because of the intensive chemical processes required to isolate fluorine and bond it to the carbon backbone. As the O-ring operates as a passive seal drawing zero energy during its operational life, this upfront energy requirement for the chemical synthesis dictates almost the entire ecological profile of the component.

7.4.2 Manufacturing Phase Burden

The next major contributor was the shaping process, requiring 0.59 MJ of energy and generating 0.0373 kg CO₂ (eq) in emissions. The primary process selected was polymer molding (*using Granta EduPack database suggestion*), which reflects the industrial injection or compression molding. While the molding process itself is quick and not too demanding, the real energy expense appears from the high-temperature post-cure bake-out stage for the aerospace and vacuum grade FKM. This step is essential to fully establish crosslinks within the network and remove volatile outgassing elements.

7.4.3 Sustainable Optimization Protocols

As already specified within the earlier sections of the document, the dynamic seal must strictly comply with the AS568-377 geometric constraints to create a successful interface with off-the-shelf flanges for SEM. Altering the cross-sectional thickness or volume to save the overall mass is thereby

not a feasible option. Therefore, environmental optimization should focus on raw material sourcing, logistics and industrial forming efficiency.

- **Low-carbon expense sourcing:** As 88.4% of the emissions stem from the raw material's chemical extraction, the manufacturer can mandate the sourcing of raw FKM gum exclusively from the chemical plants utilizing a highly decarbonized or renewable energy grid (i.e., green energy prioritized plants such as hydro-powered or nuclear-powered hubs). This would significantly drop the embedded carbon footprint of the virgin feedstock without altering the material's chemistry or geometry.
- **Flashless Tooling and Manufacturing Optimization:** Standard compression molding procedures often force excess material out of the cavity, creating a web of scrap "flash" that requires to be trimmed off and then landfilled (specifically in our case as FKM, being a thermoset cannot be remolded). Utilizing modern, high precision flashless injection molding ensures that the polymer feedstock is injected precisely into the closed cavity. This maximizes the material yield, reducing the raw material scrap rate per batch, and directly lowers down the pre-formed material mass requirement.
- **Infrared Curing:** The manufacturing phase energy can be optimized by shifting away from traditional convection batch ovens for the post-curing cycle. Implementing targeted infrared (IR) or Induction heating elements within the curing ovens allow for rapid, direct heat transfer to the highly cross-linked thermoset network chains, reducing the continuous oven runtime and electrical waste from the production line.

7.4.4 Logistical & EOL Dynamics

While the material synthesis and the manufacture stage dictate the absolute magnitude of the environmental footprint, evaluating the secondary phases confirms the efficiency of logistical and disposal parameters.

- **Transport Optimization:** As already explained, by strategically selecting international shipping by a bulk carrier rather than an intercontinental flight for 9000 km, coupled with the utilization of electric train network within Italy for 500 km regional distribution, the burden was restricted to 0.0388 MJ and 0.0028 kg CO_{2 (eq)}. The detailed breakdown shows that the maritime route comprises 77.9% of the CO₂ emission. However, the complete transport phase contributes less than 1% to the LCA.
- **Disposal Mechanics:** Because of its passive usage, the seal itself does not require any direct energy utilization during operation. When the total upfront synthesis debt is distributed over the 5-year preventive maintenance cycle, the equivalent annual environment burden is extremely low i.e., 1.52 MJ/year and 0.0698 kg CO_{2 (eq)}/year

- **EoL data:** It quantifies the physical impact of the thermoset disposal. AS FKM is a heavily cross-linked fluorinated network, inert landfilling remains the required disposal mechanism to avoid toxic incinerating byproducts. The Eco-audit tool shows that the act of landfilling is extremely passive, demanding negligible 0.0068 MJ and 0.00047 kg CO₂ (eq). So, the real cost of the EoL phase is the inability to reclaim our initial investment of 7 MJ in virgin material, thereby prompting us to extend the operational lifespan of maximizing the utility of initial chemical synthesis.

8. Conclusion

After reviewing international standards, academic literature, and utilizing GRANTA software to optimize the material selection for our specific application, we can conclude that the best approach for producing a Monolithic O-Ring for a Vacuum Seal at the HV Chamber of a SEM Device is using **FKM (Post cured by Carbon Black 20-35%)**. The dynamic elastic recovery, high-temperature performance, and mechanical compression resistance offered by the monolithic architecture are superior when compared to the composite approach for this application. Furthermore, given that the key factors for a seal in a critical high-vacuum device like the SEM are low Total Mass Loss (TML), low Compression Set, low CVCM, and Organic Solvent Resistance, this FKM formulation emerges as the material that best fulfills the requirements of the task.

Evaluating vital indexes based on maximum elastic deformation, thermo-mechanical stability, and permitted permeability through the GRANTA Software and its Ashby-based system allowed us to select a material that perfectly matches and aligns with the state of the art for this technology. Finally, with the Eco-Audit tool, we established that by adopting measures such as using a renewable-energy grid, flashless tools, infrared curing, and a well-planned route for transport and disposal throughout production, manufacturing, and treatment, we can significantly improve both the environmental and energy footprint, as well as the End-of-Life dynamics for our selected material.

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